Practical #2: Maven + Docker Integration

Aim:

To integrate Maven with Docker and run a Java application inside a Docker container.

Prerequisites:

- Java 17 installed

- Maven installed

- Docker Desktop installed and running

Steps:

1. Create Maven Project:

Command:

mvn archetype:generate -DgroupId=com.example -DartifactId=docker-maven-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

2. Navigate to Project:

cd docker-maven-demo

3. Modify Java Program:

File: src/main/java/com/example/App.java

Code:

```
package com.example;
public class App {
   public static void main(String[] args) {
      System.out.println("Hello from Docker + Maven 🚀");
   }
}
```

4. Update pom.xml:

(Add Maven Jar Plugin and Dockerfile Maven Plugin with Java version 17)

5. Build the Project:

```
mvn clean package
```

6. Create Dockerfile:

File name: Dockerfile

Content:

```
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY target/docker-maven-demo-1.0-SNAPSHOT.jar app.jar
ENTRYPOINT ["java", "-jar", "app.jar"]
```

7. Configure Docker Connection (Windows Fix):

Command:

```
set DOCKER_HOST=npipe:////./pipe/docker_engine
```

OR enable:

Docker Desktop → Settings → Enable tcp://localhost:2375

8. Build Docker Image:

```
mvn dockerfile:build
```

9. Verify Docker Image:

```
docker images
```

10. Run Docker Container:

```
docker run docker-maven-demo:1.0-SNAPSHOT
```

Output:

```
Hello from Docker + Maven 
```

Conclusion:

Successfully integrated Maven with Docker and executed a Java application inside a container.

```
C:\Users\Lenovo\docker-maven-demo>set DOCKER_HOST=npipe:////./pipe/docker_engine

C:\Users\Lenovo\docker-maven-demo>mvn dockerfile:build
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:docker-maven-demo >--------------------
[INFO] Building docker-maven-demo 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- dockerfile:1.4.13:build (default-cli) @ docker-maven-demo ---
[WARNING] 1 problem was encountered while building the effective model for org.javassist:javassist:jar:3.18.1-GA during
dependency collection step for plugin (use -X to see details)
[INFO] dockerfile: null
[INFO] contextDirectory: C:\Users\Lenovo\docker-maven-demo
[INFO] Building Docker context C:\Users\Lenovo\docker-maven-demo
[INFO] Path(dockerfile): null
[INFO] Path(contextDirectory): C:\Users\Lenovo\docker-maven-demo
[INFO]
[INFO] Image will be built as docker-maven-demo:1.0-SNAPSHOT
[INFO]
[INFO] Step 1/4 : FROM eclipse-temurin:17-jdk
[INFO]
[INFO] Pulling from library/eclipse-temurin
[INFO] Image 2124cce28a2b: Pulling fs layer
[INFO] Image 0f154401b2e6: Pulling fs layer
[INFO] Image b79217225f62: Pulling fs layer
[INFO] Image 817807f3c64e: Pulling fs layer
[INFO] Image 01efa0bb4ac6: Pulling fs layer
```

```
Command Prompt
[INFO] Status: Downloaded newer image for eclipse-temurin:17-jdk
[INFO]  ---> 2cffc61cde3d
[INFO] Step 2/4 : WORKDIR /app
[INFO]
[INFO]  ---> Running in eaaee2606879
[INFO]  ---> Removed intermediate container eaaee2606879
[INFO]  ---> 14d3fad71f73
[INFO] Step 3/4 : COPY target/docker-maven-demo-1.0-SNAPSHOT.jar app.jar
[INFO]
[INFO]  ---> 2d545aad2c8d
[INFO] Step 4/4 : ENTRYPOINT ["java", "-jar", "app.jar"]
[INFO]
[INFO]  ---> Running in 133be36a67ac
[INFO]  ---> Removed intermediate container 133be36a67ac
[INFO]  ---> 463c088eaa72
[INFO] Successfully built 463c088eaa72
[INFO] Successfully tagged docker-maven-demo:1.0-SNAPSHOT
[INFO]
[INFO] Detected build of image with id 463c088eaa72
[INFO] Building jar: C:\Users\Lenovo\docker-maven-demo\target\docker-maven-demo-1.0-SNAPSHOT-docker-info.jar
[INFO] Successfully built docker-maven-demo:1.0-SNAPSHOT
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  55.922 s
[INFO] Finished at: 2026-03-26T22:34:07+05:30
[INFO] ------------------------------------------------------------------------
```

Command Prompt

```
C:\Users\Lenovo\docker-maven-demo>docker images
```

i Info → U In Use

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
abhaychaudhary7/devops-ci-cd-app:latest	8693a6d08e15	277MB	66MB	U
abhaychaudhary7/devops-ci-cd-app:v1	8693a6d08e15	277MB	66MB	U
devops-ci-cd-app:local	380e715547cc	349MB	82MB	U
devops-project:latest	4e2ea025a202	349MB	82MB	
docker-maven-demo:1.0-SNAPSHOT	463c088eaa72	636MB	198MB	
eclipse-temurin:17-jdk	2cffc61cde3d	639MB	202MB	
flaskmulti:v1	d89062bcea60	1.6GB	411MB	U
ghcr.io/abhaychaudhary18/flask-app:v1	2a1896cd7758	212MB	52MB	U
ghcr.io/abhaychaudhary18/ghcr:v1	ef4b5e16ca2e	92.5MB	26MB	U
httpd:latest	dd178595edd6	177MB	47.6MB	U
mongo:6	03cda579c8ca	1.06GB	273MB	U
mysql:5.7	4bc6bc963e6d	700MB	149MB	U
mysql:latest	932fe8fbc04c	1.27GB	283MB	U
mywordpress:latest	0431324be098	1.07GB	266MB	
nginx:latest	c881927c4077	240MB	65.7MB	U
node-mongo-compose-app:latest	9c1de619e288	1.63GB	406MB	U
php_5215:latest	6428abbcd4c6	707MB	176MB	
php_5216:latest	0b1e50fdca35	707MB	176MB	U
python:3.11-slim	5be45dbade29	188MB	47.7MB	
pythonii:latest	05a9f90134c5	1.6GB	411MB	U
redis:latest	1c054d54ecd1	204MB	55.3MB	U
ubuntu:latest	cd1dba651b30	119MB	31.7MB	U
whaleops7/devops-ci-cd-app:v1	8693a6d08e15	277MB	66MB	U
whaleops7/devops-ci-cd-app:v2	380e715547cc	349MB	82MB	U
whaleops7/devops-ci-cd-app:vi	8693a6d08e15	277MB	66MB	U
whaleops7/nginx:v1	c881927c4077	240MB	65.7MB	U

```
C:\Users\Lenovo\docker-maven-demo>docker run docker-maven-demo:1.0-SNAPSHOT
Hello World!
```

docker desktop PERSONAL

Search Ctrl+K

A new version of Docker Desktop is available. Your current version is 73 days old. Update

Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions

Images / docker-maven-demo:1.0-SNAPSHOT

docker-maven-demo:1.0-SNAPSHOT IN USE
463c088eaa72

CREATED 25 minutes ago | SIZE 635.95 MB | Recommended fixes | Run

Analyzed by docker scout

Layers (19)

0	ARG RELEASE	0 B
1	ARG LAUNCHPAD_BUILD_ARCH	0 B
2	LABEL org.opencontainers.image.ref.name=ubuntu	0 B
3	LABEL org.opencontainers.image.version=24.04	0 B
4	ADD file:3f78aa860931e0853077f09eb31eddbeeef8a9dd...	87.6 MB
5	CMD ["/bin/bash"]	0 B
6	ENV JAVA_HOME=/opt/java/openjdk	0 B
7	ENV PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/lo...	0 B
8	ENV LANG=en_US.UTF-8 LANGUAGE=en_US:en LC_ALL=e...	0 B
9	RUN /bin/sh -c set -eux; apt-get update; DEBIAN_FRONTE...	68.81 MB
10	ENV JAVA_VERSION=jdk-17.0.18+8	0 B
11	RUN /bin/sh -c set -eux; ARCH="$(dpkg --print-architectur...	281.13 MB
12	RUN /bin/sh -c set -eux; echo "Verifying install ...."; fileEnc...	12.29 KB

Vulnerabilities | Packages

Give feedback

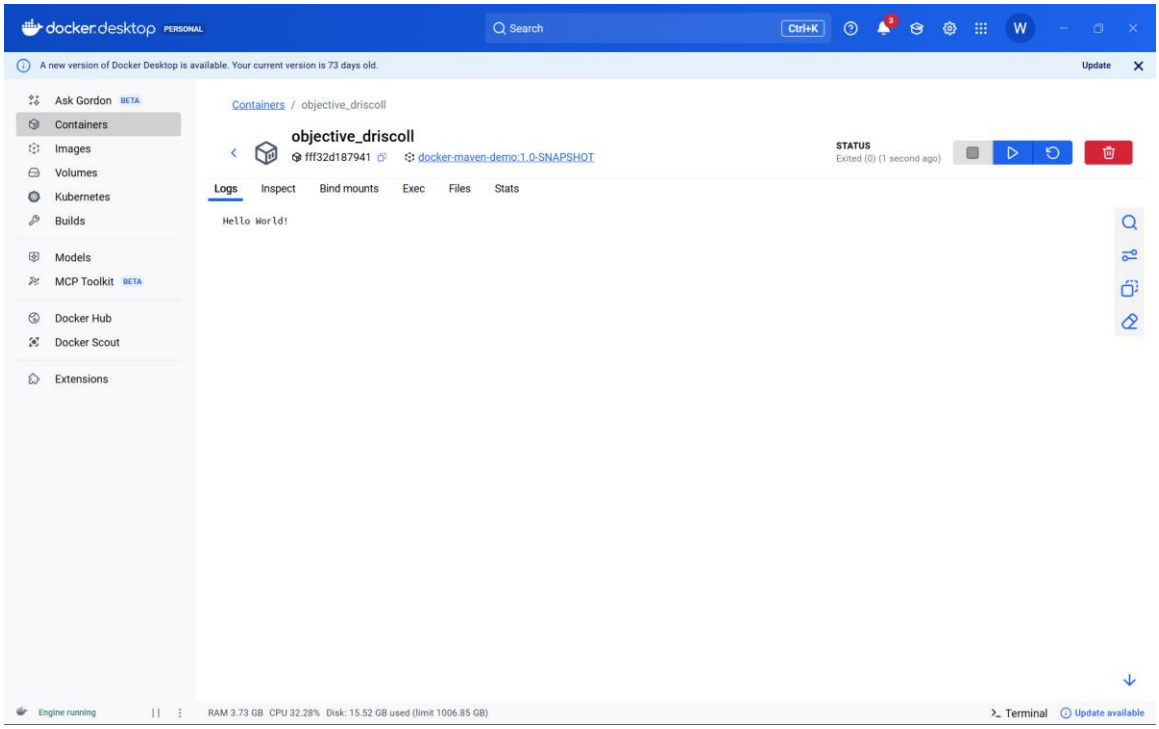
This image has not been analyzed

You can use Docker Scout to analyze local images and list its vulnerabilities.

Start analysis

Enable background indexing in Settings so your results are always ready.

Engine running | RAM 3.71 GB CPU 9.39% Disk: 15.52 GB used (limit 1006.85 GB) | Terminal | Update available


docker desktop PERSONAL
Search
Ctrl+K
A new version of Docker Desktop is available. Your current version is 73 days old.
Update
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers / objective_driscoll
objective_driscoll
fff32d187941
docker-maven-demo:1.0-SNAPSHOT
STATUS
Exited (0) (1 second ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
Hello World!
Engine running
RAM 3.73 GB CPU 32.28% Disk: 15.52 GB used (limit 1006.85 GB)
Terminal
Update available